The strands

The units are a lens; the year's courses carry the disciplines. Here is what carries each one, and where each unit draws on it.

Richard Verver, JCL · Canon Lawyer

What carries each discipline

Reading and Number, the roots of the map, run every week. So do the lower disciplines that a family's own courses already cover. These are the courses this year was built around; others of the same kind would do.

DISCIPLINE	CARRIED ALL YEAR BY	WHY IT FITS
Reading	Latin taught by Ørberg's method (*Familia Romana*), for example Schola Latina's live online classes; Art of Problem Solving's Language Arts	Ørberg lets meaning come from the Latin itself before the grammar is named: problems first again
Number	Art of Problem Solving: Beast Academy 5 or *Prealgebra*, the book and the live class	Every section opens with problems before any explanation
Nature	Science from final causes: *Caveman Chemistry*, and a school science credit where one is wanted; Big History's first thresholds	A finding fixes a fact; it never says what a thing is for
Making	Photography; *Caveman Chemistry*'s crafts	The essays' Making page names photography itself
Memory	Big History; the atrium's long ribbon of the History of the Kingdom of God	Two readings of one order of events (Unit 5)
The Person to Worship	The units, the essays and the atrium	Where most of the essays are filed

That is a full week, so the Benedict spine shrinks to fit: the morning's presentation and the unit's attention, not a separate two hours. The essays and the gap essays replace other writing rather than adding to it.

Reading and Number: Art of Problem Solving

AoPS books open every section with problems before any explanation. Richard Rusczyk's phrase for it is "We show the problems first." That is Mr. Benedict's Monday puzzle, worked before anything is explained, so the Monday puzzle can be the opening problems of the week's section.

STRAND	FOR AGES 10 TO 13	FORMAT
Number	Beast Academy Level 5 (ages 10 to 13) if it is unfinished; otherwise *Prealgebra* (over 1,000 problems). For a strong 12- or 13-year-old, *Introduction to Counting & Probability* and *Number Theory*, which AoPS places in grades 7 to 10, after algebra	The book and the live text-based class. The challenge problems stay
Reading	Honors Language Arts 5 (ages 10 to 12): historical mysteries from primary sources. Honors Language Arts 6 (ages 11 to 13): philosophy, fiction, a feature article, *A Midsummer Night's Dream*. Honors Language Arts 7 (ages 12 to 14): persuasive writing and rhetoric	A live class only, through AoPS Academy: 36 weeks, one 105-minute class a week, 60 to 90 minutes of homework, so it runs beside the year week for week

A caution about Alcumus. AoPS's practice site picks the next problem by a running measure of the child and keeps a level that can drop with a miss. For a child who reads that as a verdict on himself, leave it out: the book's problems and the class's challenge problems are enough. Measuring is fine. Letting the measure choose the next problem puts it ahead of the order, and the Number unit's own rule is that order comes first and measure is laid on it.

Where each unit draws on the roots

UNIT	FROM NUMBER	FROM READING
1 Reading	—	Case files on historical mysteries: every theory kept open until evidence rules it out. *A Midsummer Night's Dream*: readings fixed too soon
2 Number	The properties of arithmetic (what follows from what); counting (how much: 2 + 4 + 8 + 16 Morse codes)	—
3 Nature	Ratios and statistics on the noon-shadow log	—
4 Making	Geometry: the camera obscura as similar triangles	Rhetoric, which the essays' map folds into Making and Reading
5 Memory	Remainders: the day of the week of any date	Primary sources and how far to trust them
7 Being	Ratio and proportion. Euclid's word for proportion is *analogia*	—
8 God	—	Weighing a source's credibility: the unit's rank, form and date
11 Law	The three months of c. 8 §1, computed by hand	A mock trial

Number stops at the edge of The Act. In Unit 9 the eggs in Of Divers Weights run from $3.99 to $4.79, about 20% more at the top. That is computed. Whether it is fair is judged.

Latin

Ørberg's *Familia Romana* is written so that the meaning of each new word comes from the Latin around it and the pictures in the margin, before the grammar is named. It is the Reading unit's rule from the inside: every sentence is read before it is explained. Later in the year the Latin comes back on its own: the Vulgate in Unit 8, the Code of Canon Law (written in Latin) in Unit 11, and the Ordinary of the Mass in Unit 12.

Big History and photography

Big History, from the Big History Project (free from the OER Project, or in homeschool versions such as Rise Out's), runs through eight thresholds of increasing complexity, from the Big Bang to the modern revolution. It carries Nature and Memory; see Science from final causes for how to keep it in its kind.

Photography is Making with Number inside it. Each full stop of aperture or shutter halves or doubles the light: shutter speeds run 1/125, 1/250, 1/500, and f-numbers step by √2. The marked numbers are rounded measures laid on the doubling. 1/125 stands for 1/128, and f/11 for about 11.31: Unit 2's rule, inside a camera.

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